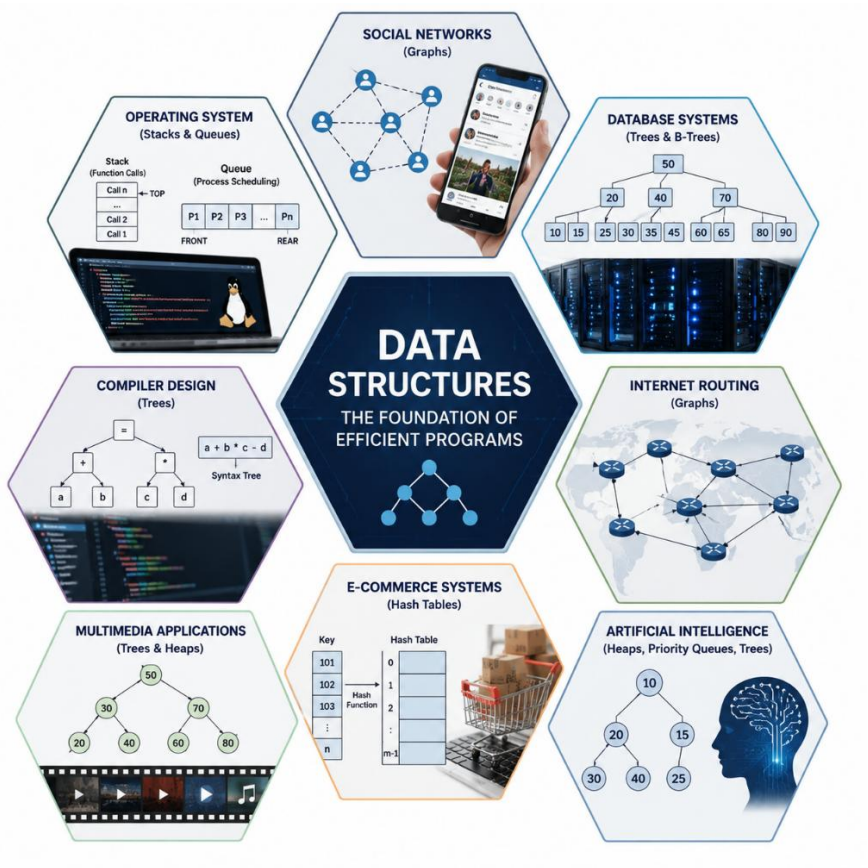
Data Structures & Algorithms Lab

ECE2132



Department of Electronics & Communication Engineering

By:

Dr. D.V. Prashant
Dr. Kamal Kishore Upadhyay

Manipal University Jaipur,

Dehmi Kalan, Off Jaipur-Ajmer Expressway, Jaipur, (Raj.)

Rajasthan 303007

MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

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EXPERIMENT NO. 2

WRITE A PROGRAM IN C++ TO IMPLEMENT POINTER ARITHMETIC.

AIM: To study and implement pointer arithmetic in C++ by performing increment, decrement and addition operations on pointer variables.

SOFTWARE USED: Visual Studio Code / Code: Blocks / Dev-C++, GCC/G++ Compiler. Operating System: Windows/Linux

THEORY: Pointer arithmetic allows arithmetic operations on pointer variables. Incrementing or decrementing a pointer moves it to the next or previous memory location based on the size of the data type. It is widely used for traversing arrays and implementing linked lists, trees, graphs and other dynamic data structures.

ALOGRITHM:

1. Start.
2. Declare an integer array.
3. Declare a pointer.
4. Point to first array element.
5. Display address and value.
6. Increment pointer.
7. Decrement pointer.
8. Add 2 to pointer.
9. Stop.

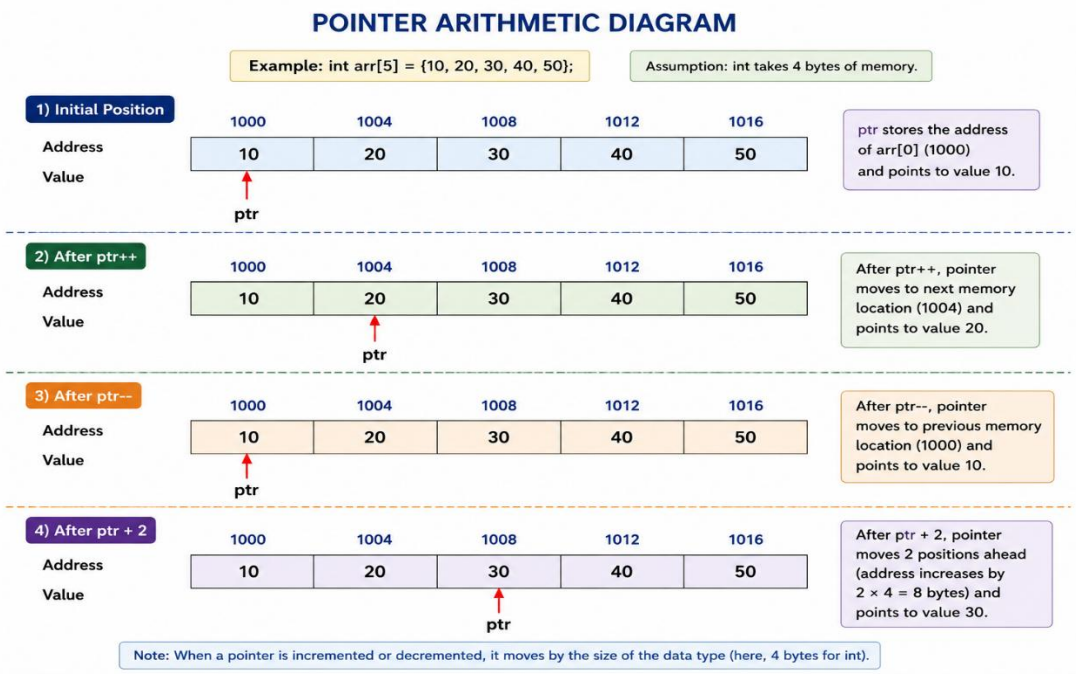


Figure.1: Pointer Movement Across Consecutive Memory Locations

PROGRAM (C++):

```
#include <iostream>
using namespace std;

int main()
{
    int arr[5]={10,20,30,40,50};
    int *ptr=arr;

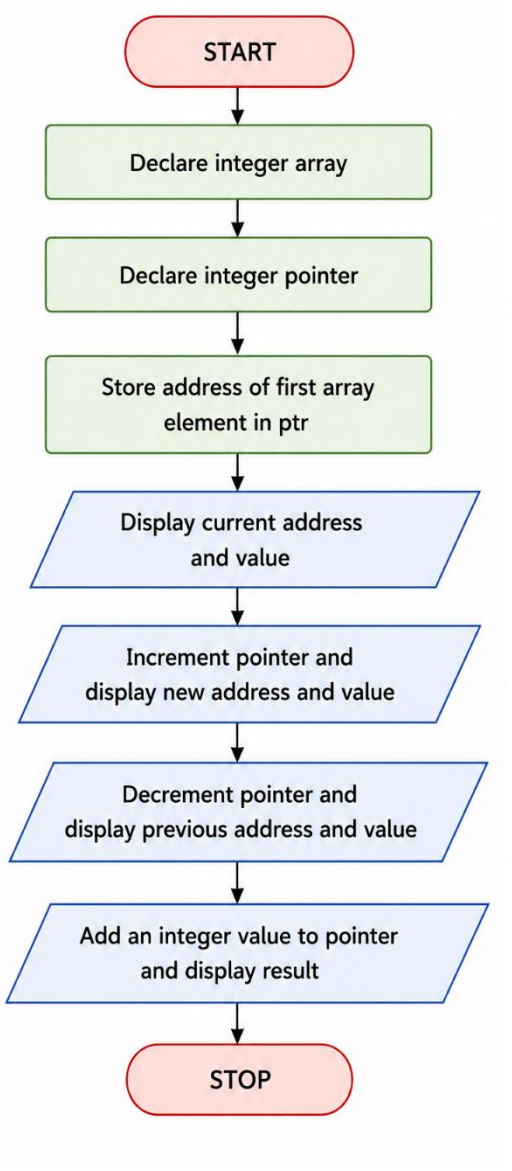
    cout<<"Initial Address : "<<ptr<<endl;
    cout<<"Value : "<<*ptr<<endl;

    ptr++;
    cout<<"\nAfter Increment"<<endl;
    cout<<"Address : "<<ptr<<endl;
    cout<<"Value : "<<*ptr<<endl;

    ptr--;
    cout<<"\nAfter Decrement"<<endl;
    cout<<"Address : "<<ptr<<endl;
    cout<<"Value : "<<*ptr<<endl;

    ptr=ptr+2;
    cout<<"\nAfter Adding 2"<<endl;
    cout<<"Address : "<<ptr<<endl;
    cout<<"Value : "<<*ptr<<endl;

    return 0;
}
```



EXPLANATION OF THE PROGRAM:

Statement	Explanation
#include <iostream>	Provides input/output functions.
int arr[5]={10,20,30,40,50};	Declares and initializes an integer array.
int *ptr=arr;	Pointer stores address of first element.
ptr++;	Moves pointer to next element.
ptr--;	Moves pointer to previous element.
ptr=ptr+2;	Moves pointer two elements forward.
*ptr	Accesses value stored at pointer address.

Sample Input: No user input is required.
Initial Address: 0x61FF00, Value: 10
After Increment, Address: 0x61FF04, Value: 20
After Decrement, Address: 0x61FF00, Value: 10
After Adding 2, Address: 0x61FF08, Value: 30
Time Complexity: O (1), Space Complexity: O (1)

Name: _____ **Reg. No.:** _____

IMPLEMENTATION OF CODE:

```cpp
/*******************************************************************************/

#include <iostream>
using namespace std;

int main()
{
    int arr[5] = {10, 20, 30, 40, 50};
    int *ptr;

    ptr = arr;

    cout << "Initial Address : " << ptr << endl;
    cout << "Value : " << *ptr << endl;

    ptr++;

    cout << "\nAfter Increment" << endl;
    cout << "Address : " << ptr << endl;
    cout << "Value : " << *ptr << endl;

    ptr--;

    cout << "\nAfter Decrement" << endl;
    cout << "Address : " << ptr << endl;
    cout << "Value : " << *ptr << endl;

    ptr = ptr + 2;

    cout << "\nAfter Adding 2" << endl;
    cout << "Address : " << ptr << endl;
    cout << "Value : " << *ptr << endl;

    return 0;
}
```

```
Initial Address : 0x7fff5ce08080
Value : 10

After Increment
Address : 0x7fff5ce08084
Value : 20

After Decrement
Address : 0x7fff5ce08080
Value : 10

After Adding 2
Address : 0x7fff5ce08088
Value : 30


...Program finished with exit code 0
Press ENTER to exit console.
```

APPLICATIONS: Array traversal, Dynamic memory allocation, Linked Lists, Trees, Graphs

RESULT: The program was successfully executed. It demonstrated how a pointer stores the address of a variable and accesses the value through dereferencing.